Research Note: LLD Practice Platform

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1. Executive Summary & Problem Space

Mastering Low-Level Design (LLD) and Object-Oriented Design (OOD) is pivotal for software engineers transitioning to building maintainable enterprise systems. However, while platforms like LeetCode have solved DSA practice with deterministic test suites, LLD practice remains fragmented and frustrating.

In LLD, there is rarely a single "correct" answer. Learners suffer from the "Echo-Chamber Effect": they draft class hierarchies or code, yet have zero visibility into whether their abstractions, decoupling, or SOLID adherences are sound.

2. Competitive Landscape & Existing Approaches

• LeetCode / HackerRank: Exceptional for DSA, but binary pass/fail ignores design patterns, encapsulation, and extensibility.
• Educative / GitHub Repos (Grokking OOD): Rich reference solutions, but passive reading only with no active sandbox or evaluation.
• Human Mock Interviews: High quality, but expensive ($150+/hr), non-scalable, and inconsistent.
• Generic Chatbots: Capable of reviewing snippets, but lack structured rubrics, contextual memory of problem requirements, and progressive attempt tracking.

3. Key Gaps in the Learner Journey

1. The Evaluation Paradox: Unit tests only verify what code outputs, not how clean or decoupled the architecture is.
2. Ambiguous Problem Scope: Learners frequently get lost in non-functional rabbit holes instead of domain modeling.
3. Absence of Iterative Feedback Loops: Learners cannot observe how their design score evolves across attempts.
4. Lack of Trade-Off Articulation: Learners are rarely prompted to explain architectural choices.

4. Product Direction & Value Proposition

Our MVP provides an interactive LLD Practice Sandbox engineered around: Practice loop -> Explainable Evaluation -> Progression Tracking.

Key features include curated classic problems (Parking Lot, Elevator, Vending Machine), a Hybrid Evaluation Architecture combining deterministic static keyword/AST rules with semantic AI reasoning, a 4-second fail-safe timeout boundary, and persistent attempt history tracking score deltas.